

Sara Alhetela

AI ENGINEER & DATA SCIENTIST

I'm an AI engineer who takes machine learning models from notebook to production — most recently an 18-classifier pipeline that's been running inside a national energy company's pre-construction workflow, cutting a days-long manual process to minutes. My work spans the full lifecycle, from data preprocessing and experiment tracking to Git-versioned, CI/CD-deployed services, and I back the research side with a peer-reviewed publication in computer vision.

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EXPERIENCE

Data Scientist

Feb 2024 - Jul 2024

Fathom.io

- 22% drop in response error rate.** Fine-tuned a Hugging Face model for task-specific performance, then built LangChain-based structured LLM evaluation rubrics and iterative prompt engineering to catch regressions before release, inside a 4-person Scrum team.
- 35% faster training cycles.** Restructured GPU cluster job scheduling and resource allocation, clearing the idle-compute bottlenecks that had been stalling weekly model releases.
- 15% uptime improvement.** Traced and resolved five recurring platform failures through root-cause analysis over the final two months of the engagement.

AI/ML Intern

Jul 2023 - Sep 2023

TAQA – Industrialization & Energy Services Company

- Days-long process cut to minutes.** Architected and deployed a 3-stage sequential pipeline (presence → variant → concentration bin) across 18 independently modeled cement additives, with configurable classifiers (Random Forest, Extra Trees, XGBoost, CatBoost, Gradient Boosting, Voting) tracked via MLflow — replacing a manual expert process that blocked every pre-construction cycle.
- 99.4% classification accuracy.** Tuned per-additive, per-stage classifier and resampling configuration (SMOTE, ADASYN, SMOTETomek, Tomek Links) to handle severe class imbalance in real well data.
- Physical lab testing rounds eliminated.** Shipped a Flask interface that put model outputs directly in front of field engineers, cutting pre-drilling turnaround time.

Independent ML Development

Aug 2024 - Present

Self-directed

- Deep RL certification (2025).** Moved from supervised, tabular ML into reinforcement learning: Bellman-equation methods, reward shaping, Double DQN, and N-step returns.
- +14.30% alpha over buy-and-hold.** Applied that study to design and open-source a Dueling Conv1D DQN trading agent, 150k training steps on AAPL OHLC data via CUDA, 70% trade win rate on held-out test data.

FLAGSHIP PROJECTS

Oil Well Cement Mix Predictor

scikit-learn · XGBoost · CatBoost · Flask · MLflow · Git

Deployed at TAQA. 3-stage sequential pipeline (presence → variant → concentration bin), 18 independently modeled additives, configurable classifier/resampling framework, 99.4% accuracy, MLflow-tracked, with a Flask front end used directly by field engineers. Full detail under Experience above.

Dueling Conv1D DQN Stock Trading Agent

PyTorch · Deep RL · CUDA · Git

Deep reinforcement learning system for algorithmic trading. Dueling Conv1D Q-Network with Double DQN and N-step returns, trained 150k steps on AAPL OHLC data via CUDA. Achieved +18.15% cumulative return on held-out test data vs +3.85% buy-and-hold (+14.30% alpha), 70% trade win rate. Modular architecture across environment simulation, model, training loop, and evaluation with buy-and-hold benchmarking.

github.com/saraalhetela/dueling-dqn-stock-trader

PUBLICATION

Deep Learning Approach to Recyclable Products Classification: Towards Sustainable Waste Management

Sustainability, vol. 15, no. 14, 2023, p. 11138 · Alhetela, Sara, et al. · peer-reviewed, CNN-based waste-stream image classifier

EDUCATION

B.Sc. Artificial Intelligence

Imam Abdulrahman Bin Faisal University

Dammam · Aug 2023

GPA 4.5 / 5.0 · First-Class Honours

SKILLS

ML / DL

PyTorch · TensorFlow

Keras · scikit-learn

XGBoost · CatBoost

LLMS & NLP

Hugging Face · Transformers

LangChain · RAG

Prompt Engineering

Fine-tuning · NLP

MLOPS & ENGINEERING

Git · MLflow · Docker

Kubernetes · CI/CD

FastAPI · Model Deployment

Unit Testing

DEEP RL

Deep Reinforcement Learning

DQN · Reward Shaping · CUDA

COMPUTER VISION

CNNs · Image Classification

DATA & CLOUD

Python · SQL · NumPy

Pandas · BigQuery ML · GCP

Data Engineering

LANGUAGES

Arabic Native

English Advanced

French Intermediate

CERTIFICATIONS

Udemy

– Deep RL using Python 2025

Google Cloud Skills Boost

– Data Engineering 2021

– Machine Learning APIs 2021

Kaggle

– Feature Engineering 2023

– Intro to Machine Learning 2023